# DS701 Homework 7: Least Squares, Logistic Loss and Regularization

Released Mon Oct 19 · due Mon Oct 26, 11:59 pm on Gradescope

**Name:** _____ **BU ID:** _____

**How this works.** Answer **by hand on paper** (or a tablet with a stylus), show your work, then scan and upload to Gradescope, matching each problem to its pages. This is deliberate practice for the closed-notes quizzes, which are in the same style — so do it as much as possible without a computer or an AI assistant. If you are stuck, you can try AI in tutorial style or reach out to course staff. Feel free to use a calculator for arithmetic. For the quiz, the problems will be simplified to avoid tedious calculations. It is **participation-graded**: a serious attempt at every problem earns full credit, and worked solutions are released the day after the deadline. Budget 1–2 hours.

**Covers:** Lecture 10 (Linear Regression), Lecture 11A (Logistic Regression, Mon Oct 19) and Lecture 11B (Regularization, Wed Oct 21). Problem 3 uses Lecture 11B.

## Problem 1: Least squares by hand, and as a projection

*Practices: forming and solving the normal equations, and reading the result geometrically.*

Fit the line $y = \beta_0 + \beta_1 x$ to the four points

$$(x, y) \in \{(1, 2),\ (2, 3),\ (3, 5),\ (4, 6)\}.$$

(a) Write the design matrix $X$ (with a column of ones for the intercept) and the vector $\mathbf{y}$. Form $X^T X$ and $X^T \mathbf{y}$, and solve the normal equations $X^T X \beta = X^T \mathbf{y}$ for $\hat{\beta}_0, \hat{\beta}_1$. This produces a system of 2 equations in 2 unknowns thatyou can solve directly without matrix inversion. (Both are exact decimals — no rounding needed.)

$\hat{\beta}_0 =$ _____ $\hat{\beta}_1 =$ _____

(b) Compute the fitted values $\hat{\mathbf{y}} = X\hat{\beta}$ and the residual vector $\mathbf{r} = \mathbf{y} - \hat{\mathbf{y}}$. Verify $X^T \mathbf{r} = \mathbf{0}$ — i.e. that $\mathbf{r}$ is orthogonal to *both* columns of $X$. Then compute TSS, RSS and $R^2$.

$\hat{\mathbf{y}} =$ _____ $\mathbf{r} =$ _____ TSS = _____
RSS = _____ $R^2 =$ _____

(c) In one or two sentences: the normal equations express "residual $\perp$ column space of $X$". Explain what "$\hat{\mathbf{y}}$ is the projection of $\mathbf{y}$ onto the column space of $X$" means for these four points — which set is $\mathbf{y}$ being projected onto, and why the projection is the least-squares choice.

(d) Suppose a colleague accidentally adds the feature $x$ a second time, so the design matrix becomes $X' = [\mathbf{1} \quad \mathbf{x} \quad \mathbf{x}]$ $(4 \times 3)$. Show that $X'^T X'$ is singular (a one-line argument or a determinant will do). What can you say about the least-squares coefficient vector now — is it unique? — and about the fitted values $\hat{\mathbf{y}}$? Give two different coefficient vectors that produce the same fit as in (a).

Two coefficient vectors $(\beta_0, \beta_1, \beta_2)$: _____ and _____

## Problem 2: Logistic loss and one gradient step

*Practices: the sigmoid, the log-loss, and the chain rule that makes its gradient so simple.*

Consider a one-feature logistic model $p(y{=}1 \mid x) = \sigma(wx + b)$ with $\sigma(z) = 1/(1 + e^{-z})$. Current parameters: $w = 1$, $b = -1$. Three labelled points:

$$(x, y) \in \{(2, 1),\ (0, 0),\ (-1, 1)\}.$$

The **log-loss** (negative log-likelihood) of one example is $\ell_i = -[y_i \log p_i + (1 - y_i) \log(1 - p_i)]$ with $p_i = \sigma(wx_i + b)$; the total loss is $\ell = \sum_i \ell_i$. Minimizing $\ell$ is the same as maximizing the likelihood $L$. You may use $e^{-1} \approx 0.368$, $e^{-2} \approx 0.135$, $\ln 0.731 \approx -0.313$, $\ln 0.119 \approx -2.127$, and $\sigma'(z) = \sigma(z)(1 - \sigma(z))$ (take it as given, or verify it in one line).

(a) For each point compute $z_i = wx_i + b$, $p_i = \sigma(z_i)$ (3 decimals) and $\ell_i$. Report the total loss $\ell$. Which point is the model most wrong about? You can use a calculator to compute the sigmoid and log-loss. Problems like this will be simplified for the quiz.

| $x_i$ | $y_i$ | $z_i$ | $p_i$ | $\ell_i$ |
|---|---|---|---|---|
| 2 | 1 | _____ | _____ | _____ |
| 0 | 0 | _____ | _____ | _____ |
| $-1$ | 1 | _____ | _____ | _____ |

Total $\ell =$ _____ Most wrong about the point with $x =$ _____

(b) Show, for a single example, that $\dfrac{\partial \ell_i}{\partial w} = (p_i - y_i)\, x_i$ and $\dfrac{\partial \ell_i}{\partial b} = p_i - y_i$. (Chain rule: $\ell_i \to p_i \to z_i \to w$; the given $\sigma'$ does the heavy lifting.)

(c) Using (b), compute $\partial \ell / \partial w$ and $\partial \ell / \partial b$ summed over the three points, and take one gradient-**descent** step with learning rate $\alpha = 0.5$: $w \leftarrow w - \alpha\, \partial \ell / \partial w$, $b \leftarrow b - \alpha\, \partial \ell / \partial b$. Report the new $(w, b)$ to 3 decimals. In which direction did $b$ move, and why does that make sense given (a)?

$\partial \ell / \partial w =$ _____ $\partial \ell / \partial b =$ _____ new $(w, b) =$ _____

(d) (Optional) A classmate proposes fitting the same model by minimizing squared error $\sum_i (p_i - y_i)^2$ instead. Give two reasons, in a short paragraph, why the lecture fits logistic regression by maximum likelihood (i.e. minimizes the log-loss) rather than squared error. Hint: what does the least-squares criterion assume about the noise, and how large a penalty does each loss assign to a confident wrong prediction such as $p_i = 0.99$ when $y_i = 0$?

## Problem 3: Ridge shrinks, LASSO zeroes

*Practices: what the L2 and L1 penalties actually do to a coefficient, and how to choose the penalty weight.*

Throughout, $\lambda \geq 0$ is the penalty weight.

(a) **Ridge in one dimension.** For a no-intercept model $y \approx \beta x$ with data $(x, y) \in \{(1,1),\ (2,4),\ (3,7)\}$, the ridge objective is $J(\beta) = \sum_i (y_i - \beta x_i)^2 + \lambda\beta^2$. By setting $dJ/d\beta = 0$, show that $\hat{\beta}_\lambda = \dfrac{\sum_i x_i y_i}{\sum_i x_i^2 + \lambda}$ (this is $(X^T X + \lambda I)^{-1} X^T \mathbf{y}$ with $X$ a single column). Evaluate it for $\lambda = 0, 1, 10$ (3 decimals) and describe in a sentence what happens as $\lambda$ grows. Can any finite $\lambda$ make $\hat{\beta}_\lambda$ exactly 0?

$\hat{\beta}_0 =$ _____ $\hat{\beta}_1 =$ _____ $\hat{\beta}_{10} =$ _____ exactly 0? _____

(b) **LASSO in one dimension.** The building block of the LASSO is the scalar problem

$$\min_{\beta} \ \tfrac{1}{2}(\beta - z)^2 + \lambda|\beta|,$$

where $z$ is the unpenalized (least-squares) value. Assume first $z > 0$. Show that the minimizer is $\widehat{\beta} = z - \lambda$ if $z > \lambda$ and $\widehat{\beta} = 0$ if $z \le \lambda$. (Argue in two cases: if the minimizer is $\beta > 0$, differentiate; if $\beta < 0$, show it cannot be a minimizer when $z > 0$; the remaining candidate is $\beta = 0$.) By symmetry the general answer is $\widehat{\beta} = \text{sign}(z)\max(|z| - \lambda, 0)$ — the *soft threshold*. Evaluate it for $\lambda = 1$ and $z = 3$, $z = 0.5$, $z = -2$.

LASSO, $\lambda = 1$: $z = 3 \to$ _____ $z = 0.5 \to$ _____ $z = -2 \to$ _____

(c) The ridge version of the same scalar problem, $\frac{1}{2}(\beta - z)^2 + \lambda\beta^2$, has minimizer $\widehat{\beta} = z/(1 + 2\lambda)$ (you need not derive it). Evaluate it for $\lambda = 1$ and the same three $z$, and then explain in two or three sentences, using what you found in (a)–(c), why the L1 penalty produces coefficients that are *exactly* zero while the L2 penalty only shrinks them. (Think about the slope of $|\beta|$ versus $\beta^2$ at $\beta = 0$.)

Ridge, $\lambda = 1$: $z = 3 \to$ _____ $z = 0.5 \to$ _____ $z = -2 \to$ _____

(d) **Scenario.** An analyst fits LASSO to predict house price from `sqft` (values in the thousands), `bedrooms` (1–5) and `age_years` (0–100), without standardizing, and chooses $\lambda$ as the value that minimizes the *training* RSS. State the two things wrong with this procedure and what to do instead (one or two sentences each).